OFFICIAL ANSWER KEY & MARKING SCHEME

Physics Midterm · Max Marks: 20

Q1 (4 Marks):

- Law: The rate of change of momentum is directly proportional to applied force (2 marks).
- Mathematical: F = dp/dt = d(mv)/dt = m(dv/dt) = ma (2 marks).

Q2 (6 Marks):

- Part (a): a = (v - u)/t = (20 - 0)/10 = 2 m/s^2 (3 marks).
- Part (b): F = m * a = 1500 kg * 2 m/s^2 = 3000 N (3 marks). Unit 'N' is mandatory.

Q3 (5 Marks):

- Definition: Work done is dot product of force and displacement, W = F * s (2 marks).
- Calculation: W = 50 N * 6 m = 300 Joules (J) (3 marks).

Q4 (5 Marks):

- Kinetic Energy: Energy possessed due to motion (e.g., bullet fired from gun) (2.5 marks).
- Potential Energy: Energy stored due to configuration/position (e.g., compressed spring) (2.5 marks).